

END-TO-END DATA ANALYTICS CASE STUDY

Final Project Report

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Internship: Data Analytics Internship
Duration: 4 Weeks
Date: July 17, 2026

Tools Used

Microsoft Excel | MySQL | Python (Pandas) | Power BI

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1. Introduction

Project Overview

This internship project focused on developing practical, job-ready skills in data analytics through a structured four-week learning program. Across the four weeks, a range of industry-standard tools and techniques were applied to clean, analyze, visualize, and interpret real business data drawn from two datasets: the Microsoft Financial Sample dataset and the Global Superstore 2016 dataset.

The project progressed logically from spreadsheet-based analysis (Week 1), to database querying with SQL (Week 2), to programmatic data wrangling and visualization with Python (Week 3), and finally to interactive business intelligence dashboarding with Power BI (Week 4). Each week built on the skills of the previous one, culminating in a complete, end-to-end analytics workflow — from raw data to stakeholder-ready insight.

Objective

The objective of this capstone report is to document and demonstrate the complete analytical process undertaken during the internship, including data preparation, exploratory analysis, visualization, and business reporting, and to summarize the key findings and recommendations that emerged from the work.

2. Week 1 – Excel Dashboard

Objective

The objective of Week 1 was to learn how to analyze business data using Microsoft Excel by building Pivot Tables, Pivot Charts, and a fully interactive dashboard.

Dataset

Global Superstore 2016 — a global retail transactions dataset containing 51,290 order line items across Furniture, Office Supplies, and Technology categories, spanning five markets (USCA, Europe, Asia Pacific, LATAM, Africa) and the years 2012–2015.

Activities Performed

The sales dataset was imported into Microsoft Excel and multiple Pivot Tables were created to summarize key business metrics. A single-page interactive dashboard was then assembled from these pivots, combining KPI callouts, charts, and slicers into one view.

Dashboard Components

- Total Revenue KPI card
- Sales by Category (bar chart)
- Yearly Sales Trends (line chart, 2012–2015)
- Department-wise Revenue (horizontal bar chart, 15 sub-categories)

Interactive Slicers

Slicers were added to allow the dashboard to be filtered dynamically by:

- Market
- Region
- Category
- Segment
- Year
- Ship Mode

These slicers give end users the ability to explore sales performance from multiple business perspectives without needing to alter the underlying pivot tables.

Tools Used

- Microsoft Excel
- Pivot Tables
- Pivot Charts
- Slicers

Excel Dashboard Screenshot

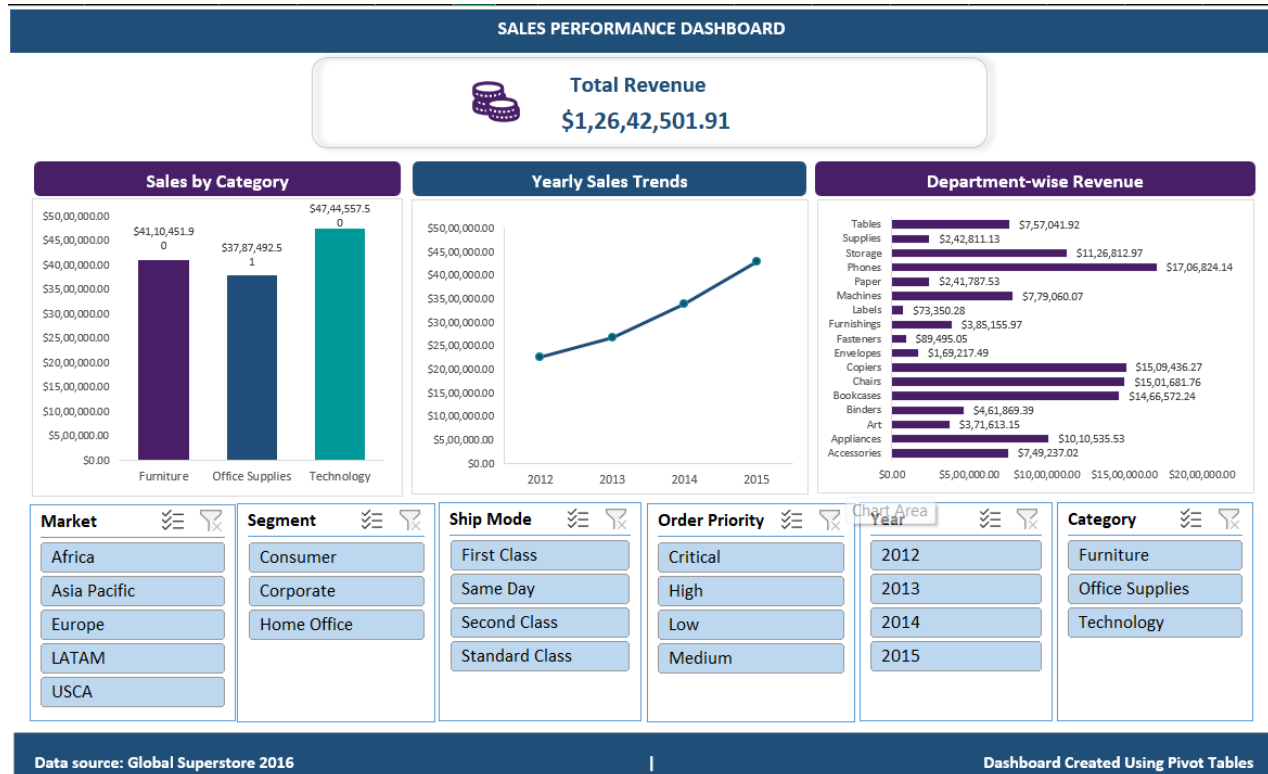


Figure 2.1 — Sales Performance Dashboard built in Microsoft Excel using Pivot Tables and Slicers

Key Metrics from the Dashboard

\$1,26,42,501.91 Total Revenue	3 Product Categories	5 Global Markets	4 Years Covered (2012–15)
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The dashboard shows Technology as the leading category by revenue (\$47,44,557.50), narrowly ahead of Furniture (\$41,10,451.90) and Office Supplies (\$37,87,492.50). The Yearly Sales Trend confirms consistent year-over-year growth from 2012 through 2015, with the steepest increase occurring between 2014 and 2015. At the department level, Phones, Chairs, Storage, and Copiers stand out as the highest-revenue sub-categories.

3. Week 2 – SQL for Data Analysis

Objective

The objective of Week 2 was to perform business analysis using SQL queries and to understand how relational databases are used to retrieve and summarize meaningful business information.

Activities Performed

A sales database (SalesDB) was created, and the dataset was imported into a table named sales. A structured sequence of SQL queries was then written and executed to progressively answer business questions, moving from simple row retrieval to aggregation, conditional classification, subqueries, and joins.

SQL Concepts Practiced

- SELECT and WHERE filtering
- ORDER BY (ascending and descending)
- GROUP BY with SUM(), COUNT(), AVG()
- CASE statements for conditional classification
- Subqueries (scalar and correlated)
- HAVING clause for filtered aggregates
- Self JOIN operations
- LIMIT for top-N analysis

Representative Queries

Revenue by category:

```
SELECT category,  
       SUM(total_price) AS Total_Sales  
FROM sales  
GROUP BY category;
```

Top 10 customers by total spend:

```
SELECT customer_name,  
       SUM(total_price) AS Total_Spent  
FROM sales  
GROUP BY customer_name  
ORDER BY Total_Spent DESC  
LIMIT 10;
```

Order value classification using CASE:

```
SELECT  
CASE  
  WHEN total_price >= 1000 THEN 'High Value'  
  WHEN total_price >= 500 THEN 'Medium Value'  
  ELSE 'Low Value'  
END AS Order_Category,  
COUNT(*) AS Total_Orders
```

```
FROM sales
GROUP BY Order_Category;
```

Customers whose spend exceeds the overall average order value (subquery + HAVING):

```
SELECT customer_name,
       SUM(total_price) AS Total_Spent
FROM sales
GROUP BY customer_name
HAVING SUM(total_price) >
(
  SELECT AVG(total_price)
  FROM sales
);
```

Business Questions Answered

- Total Revenue and Average Order Value
- Top 10 Customers and Top 10 Products by revenue
- Revenue by Category and by Region
- Order-level value segmentation (High / Medium / Low)

Query Results Summary

Executing the query set above against the sales data produced the following headline figures:

\$1,26,42,502 Total Revenue	\$246.49 Avg. Order Value	25,728 Total Orders	3.48 Avg. Quantity/Order
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Revenue by Category

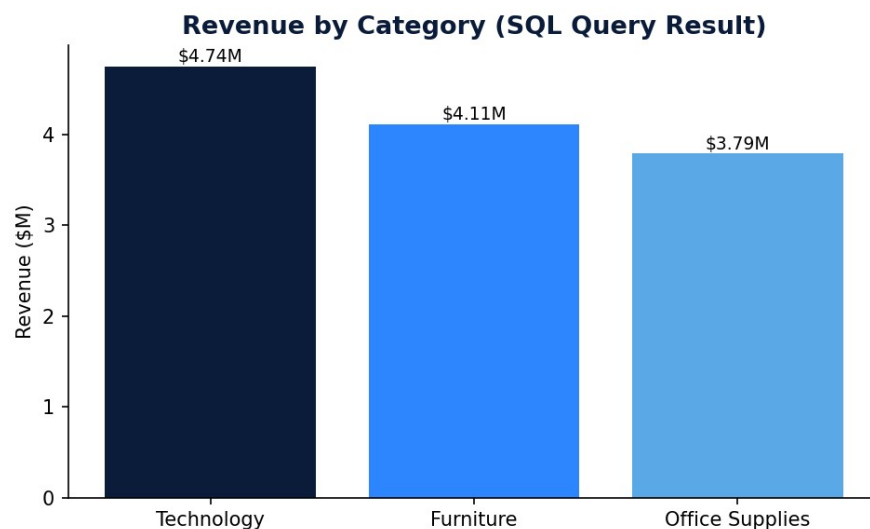


Figure 3.1 — Revenue by Category, generated from the GROUP BY category query

Top Regions by Revenue

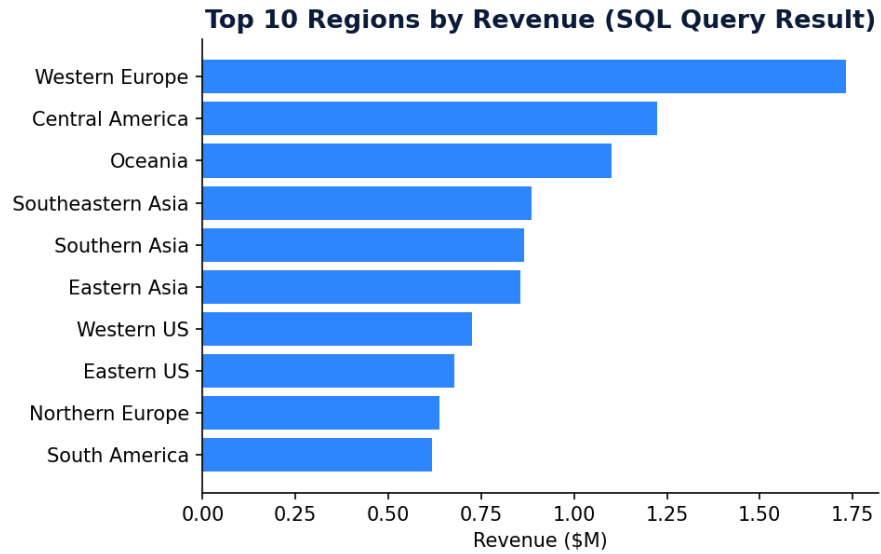


Figure 3.2 — Top 10 regions by revenue, generated from the `GROUP BY` region query

Top Customers by Spend

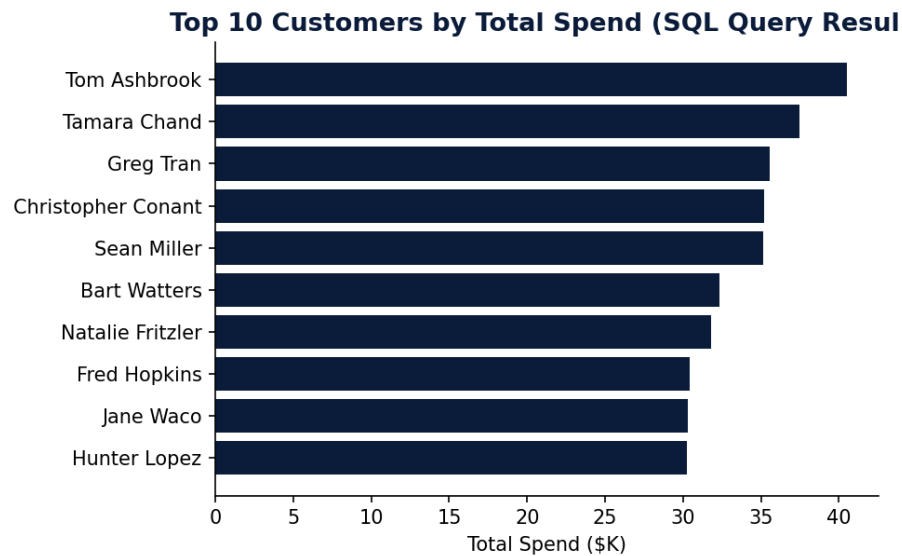


Figure 3.3 — Top 10 customers by total spend (`LIMIT 10` query)

Order Value Classification

Order Value Classification (CASE Statement)

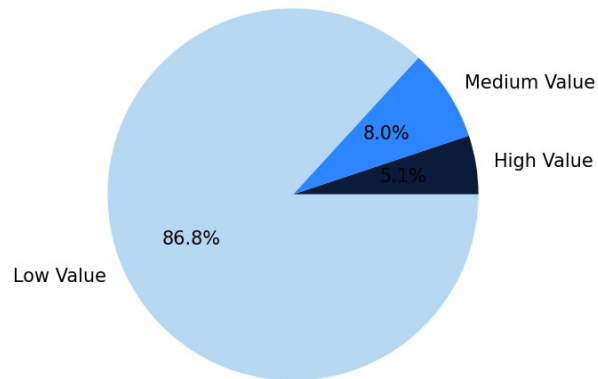


Figure 3.4 — Distribution of orders by value tier, generated from the CASE statement query

The classification shows that the large majority of individual order line items (roughly 87%) fall into the Low Value tier (under \$500), while High Value orders (over \$1,000) — though only about 5% of line items — contribute disproportionately to total revenue, underscoring the importance of large-account retention.

Tools Used

- MySQL
- MySQL Workbench

4. Week 3 – Python & Data Wrangling

Objective

The objective of Week 3 was to clean and prepare raw data using Python and to perform exploratory data visualization using Matplotlib and Seaborn.

Dataset

Workout Dataset — a small fitness-tracking dataset (32 records) containing Duration, Date, Pulse, Maxpulse, and Calories fields, used to practice the full data-cleaning workflow on a manageable scale before applying it to larger datasets.

Activities Performed

Step 1 – Load and Inspect the Dataset

```
df = pd.read_csv("data.csv")
df.head()
```

The dataset was loaded into a Pandas DataFrame containing 32 rows and 5 columns (Duration, Date, Pulse, Maxpulse, Calories).

Step 2 – Explore the Dataset

```
df.shape      # (32, 5)
df.info()
df.describe()
```

The exploratory summary showed an average workout duration of 68.4 minutes, an average pulse of 103.5 bpm, and an average calorie burn of 304.7, with a maximum single-session burn of 479.0 calories.

Step 3–4 – Identify and Handle Missing Values

```
df["Calories"] = df["Calories"].fillna(df["Calories"].mean())
df.dropna(subset=["Date"], inplace=True)
```

Two missing values were found in the Calories column and were imputed with the column mean; one row with a missing Date was removed entirely.

Step 5 – Remove Duplicate Records

```
df.drop_duplicates(inplace=True)
df.shape      # (30, 5)
```

Step 6 – Filter the Dataset

```
high_pulse = df[df["Pulse"] > 110]
```

Filtering identified all workout sessions where the pulse exceeded 110 bpm, isolating higher-intensity sessions for further review.

Step 7 – Create a New Column

```
df["Intensity"] = df["Pulse"].apply(
    lambda x: "High" if x > 120 else "Normal"
)
```

A derived Intensity column was engineered to classify each session as High or Normal based on pulse rate, adding an analytical dimension not present in the raw data.

Step 8 – Statistical Summary

```
print("Average Calories:", df["Calories"].mean())    # 307.24
print("Average Pulse:", df["Pulse"].mean())          # 103.73
print("Maximum Calories:", df["Calories"].max())     # 479.0
print("Minimum Calories:", df["Calories"].min())     # 195.1
```

Step 9–12 – Data Visualization

Four visualizations were produced to explore relationships within the cleaned dataset:

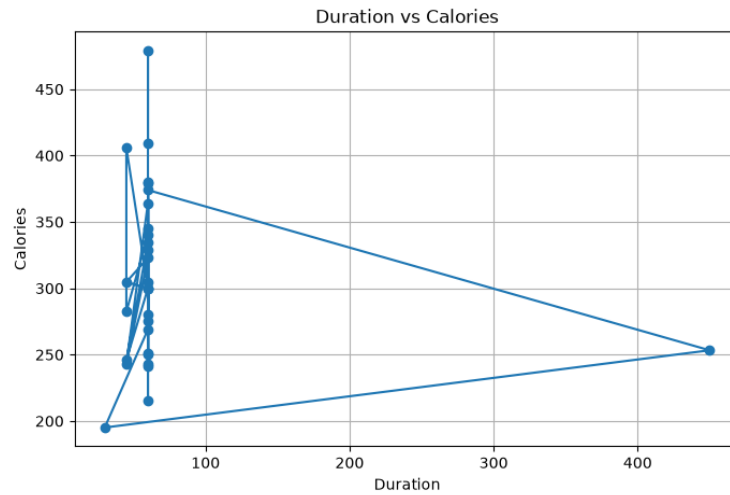


Figure 4.1 — Line chart: Duration vs. Calories

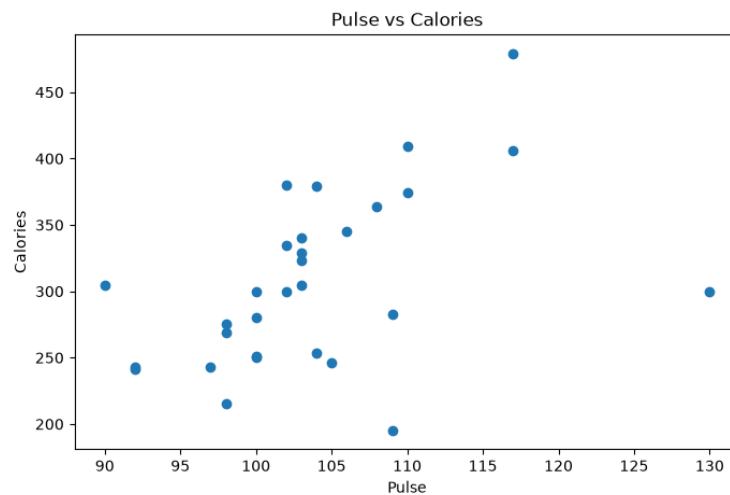


Figure 4.2 — Scatter plot: Pulse vs. Calories

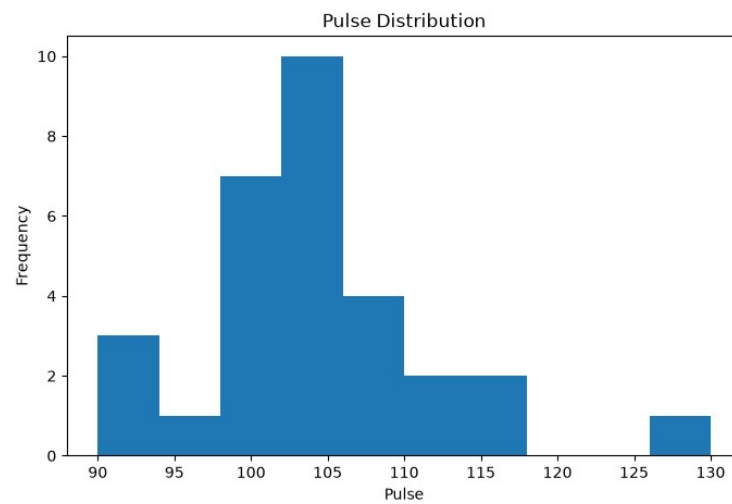


Figure 4.3 — Histogram: Pulse distribution

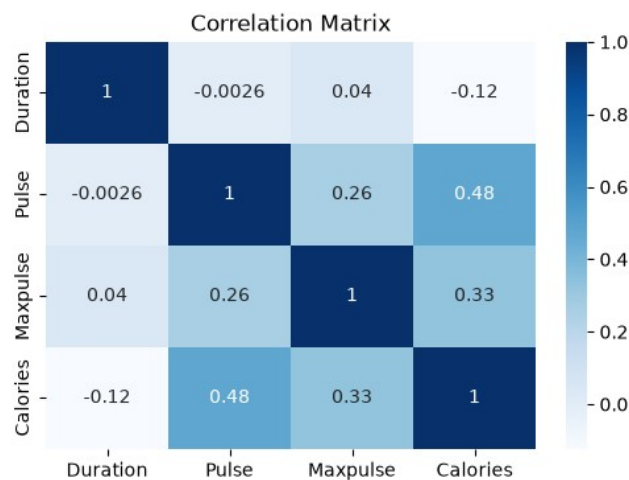


Figure 4.4 — Correlation heatmap across all numeric fields

The heatmap confirms a strong positive correlation between Duration and Calories burned, and a moderate positive correlation between Pulse and Calories, consistent with expected exercise physiology.

Step 13 – Save the Cleaned Dataset

```
df.to_csv("cleaned_data.csv", index=False)
```

Libraries Used

- Pandas — data loading, cleaning, and transformation
- Matplotlib — line, scatter, and histogram plots
- Seaborn — correlation heatmap

5. Week 4 – Power BI Dashboard

Objective

The objective of the final week was to build an interactive business intelligence dashboard using Microsoft Power BI, consolidating financial KPIs into a single stakeholder-facing view.

Dataset

Microsoft Financial Sample dataset — 700 transaction records covering five countries, five customer segments, and six products across FY2013–FY2014.

Activities Performed

The financial dataset was imported into Power BI and transformed into a fully interactive dashboard using Power Query and native Power BI visuals.

KPI Cards

- Total Sales
- Total Profit
- Units Sold
- Total Discounts
- Profit Margin

Visuals Included

- Sales by Country (bar chart)
- Profit by Product (column chart)
- Monthly Sales Trend (line chart)
- Sales by Segment (donut chart)
- Profit vs. Sales (scatter plot)
- Sales by Discount Band (column chart)

Interactive Slicers

- Country
- Product
- Segment
- Year

Power BI Dashboard Screenshot



Figure 5.1 — Financial Performance Dashboard built in Power BI

Key Metrics from the Dashboard

\$118.73M Total Sales	\$16.89M Total Profit	1.13M Units Sold	14.2% Profit Margin
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The dashboard reveals that sales nearly tripled year-over-year (from \$26.4M in 2013 to \$92.3M in 2014), Paseo emerged as the top profit-generating product, and the Government and Small Business segments together account for roughly 80% of total revenue. Full analysis of these findings, including strategic recommendations, was delivered separately as a dedicated stakeholder insights summary.

Tools Used

- Microsoft Power BI

6. Overall Findings

Based on the analysis performed across all four weeks of the internship, the following cross-cutting observations were made:

- Interactive dashboards (Excel and Power BI) make complex, multi-dimensional business data significantly easier for non-technical stakeholders to understand and act on.
- SQL is highly effective for querying, aggregating, and segmenting large transactional datasets directly at the source, reducing the need to move data into other tools for basic analysis.
- Python and Pandas simplify data cleaning and preprocessing tasks — handling missing values, duplicates, and derived columns — that would be cumbersome to perform manually in a spreadsheet.
- Power BI provides richer interactivity and cross-filtering than static Excel dashboards, making it better suited for ongoing stakeholder reporting.
- Combining these four tools in sequence — Excel, SQL, Python, and Power BI — produces a complete, end-to-end analytics workflow: from raw data ingestion through cleaning, querying, and final visualization.

7. Recommendations

Based on the experience and results gained during this project, the following recommendations are suggested for future analytics work:

- Continue using interactive dashboards to monitor business performance on a regular (weekly or monthly) cadence.
- Establish a standard data-cleaning checklist (missing values, duplicates, type validation) before any analysis begins, following the Week 3 Python workflow.
- Use SQL as the first line of analysis for large datasets, reserving Python for tasks that require more advanced transformation or statistical modeling.
- Adopt Power BI as the primary tool for stakeholder-facing presentations and recurring business reporting, given its superior interactivity.
- Refresh underlying datasets on a regular schedule to ensure dashboards and reports continue to reflect current business conditions.

8. Conclusion

This four-week internship provided practical, hands-on experience across the complete data analytics lifecycle. Each week introduced a distinct tool and demonstrated its specific role in solving real business problems: Microsoft Excel for rapid dashboard creation, SQL for structured data querying, Python for data cleaning and exploratory visualization, and Power BI for polished, interactive business intelligence reporting.

Together, these four tools provided a comprehensive, practical understanding of the modern data analytics workflow — from raw, messy data to clean, structured insight, to a final, decision-ready dashboard. The knowledge and skills developed during this internship have strengthened both technical proficiency and the ability to analyze, visualize, and communicate business insights effectively to stakeholders.

References

- Python Documentation — <https://docs.python.org>
- Pandas Documentation — <https://pandas.pydata.org/docs>
- Matplotlib Documentation — <https://matplotlib.org>
- Seaborn Documentation — <https://seaborn.pydata.org>
- MySQL Documentation — <https://dev.mysql.com/doc>
- Microsoft Excel — <https://support.microsoft.com/excel>
- Microsoft Power BI — <https://learn.microsoft.com/power-bi>
- Global Superstore 2016 dataset; Microsoft Financial Sample dataset

Project Links

GitHub Repository: github.com/IshikaPrajapati/End-to-End-Analysis

LinkedIn Profile: [linkedin.com/in/ishika-prajapati](https://www.linkedin.com/in/ishika-prajapati)

End of Report